Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · DIFFERENCE-IN-DIFFERENCES (DID)

policy effect, before/after × treated/control

Table. DiD model

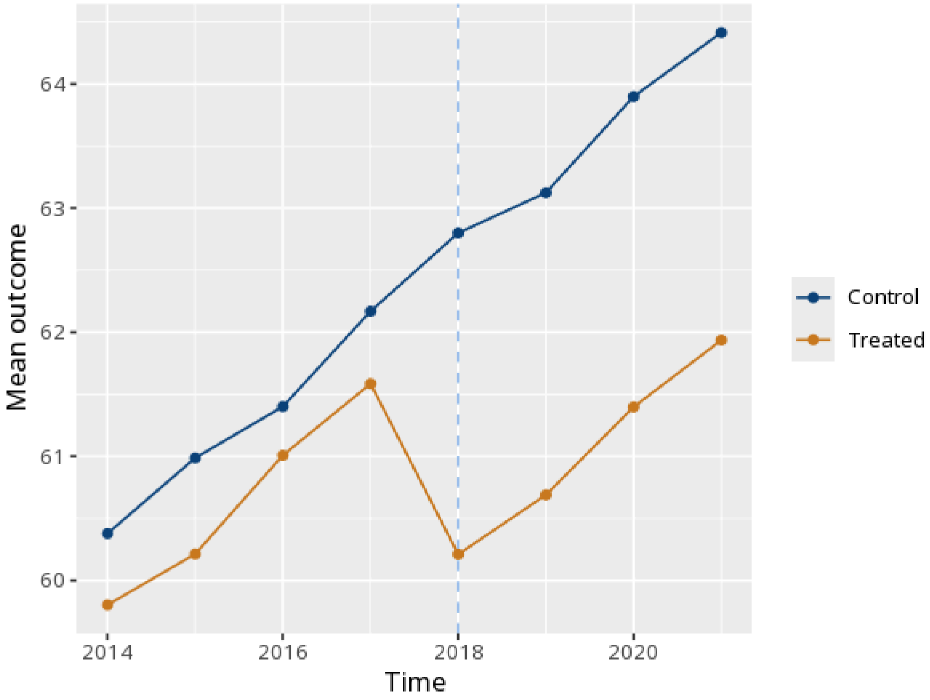
	(1)
Post	2.33
	(0.04)
	[2.24, 2.41]
Treated × Post	−1.92
	(0.08)
	[−2.07, −1.76]
Num.Obs.	160
N entities	20
Within R^2	.59
F	$F(2, 138) = 98.41$, $p < .001$

Table 2. Group × period means

Group	Pre	Post
Control	61.23	63.56
Treated	60.65	61.06

the Treated×Post interaction is the DiD effect; valid only if pre-treatment trends are parallel. Under entity fixed effects the time-invariant Treated main effect is absorbed (only Post and Treated×Post are estimated). Standard errors in parentheses are clustered by entity; the bracketed line is the 95% CI. Pre-trends test (pre-period leads-and-lags joint F of treated×time): $F(3, 72) = 0.01$, $p = .998$ — a small p flags diverging pre-trends.

Figure. Parallel trends



Understanding the numbers

Treated×Post (B). The estimated causal effect of the treatment, holding entity and period fixed effects constant. Here, B = −1.92, 95% CI [−2.07, −1.76], $p < .001$.

N. The number of observations (entity-period rows) used to fit the model. Here, N = 160.

N entities. How many distinct units (e.g. states, firms) are represented, each contributing its own fixed effect. Here, N entities = 20.

Within R^2. The share of within-entity variance (after removing each entity's own average) explained by the model. Here, within $R^2 = .59$.

F. The overall model F-test for the within regression. Here, the overall model test is $F(2, 138) = 98.41$, $p < .001$.

Pre. The raw (unadjusted) mean outcome for this group before treatment. Here, the Control group's pre-period mean is 61.23.

Post. The raw (unadjusted) mean outcome for this group after treatment. Here, the Control group's post-period mean is 63.56.

How to read this test

The Treated×Post coefficient is the estimated treatment effect. It rests on the parallel-trends assumption: that the groups would have moved together absent treatment. Similar pre-treatment trends (inspect the pre-period of the plot) make this more plausible but do not prove it — a visual check is supportive, not confirmatory, since the assumption is about the unobservable post-period counterfactual. The note adds a formal pre-trends test (a pre-period leads-and-lags joint F of the treated×time interactions): a small p flags diverging pre-trends, while a large p is consistent with parallel trends.

Method: R plm package with clustered standard errors (Croissant & Millo, 2008; Bertrand, Duflo & Mullainathan, 2004). Your significance threshold (α) is 0.05.

APA template: The DiD estimate was B=−1.92, 95% CI [−2.07, −1.76], p < .001 (clustered SE).

Statistical basis: Difference-in-differences via entity fixed effects (plm within estimator) (Croissant, Y., Millo, G. (2008). "Panel Data Econometrics in R: The plm Package." Journal of Statistical Software, 27(2), 1-43.); Clustered standard errors / parallel-trends interpretation (Bertrand, M., Duflo, E., Mullainathan, S. (2004). "How Much Should We Trust Differences-in-Differences Estimates?" Quarterly Journal of Economics, 119(1), 249-275.). Full references in the exported CITATIONS.txt.